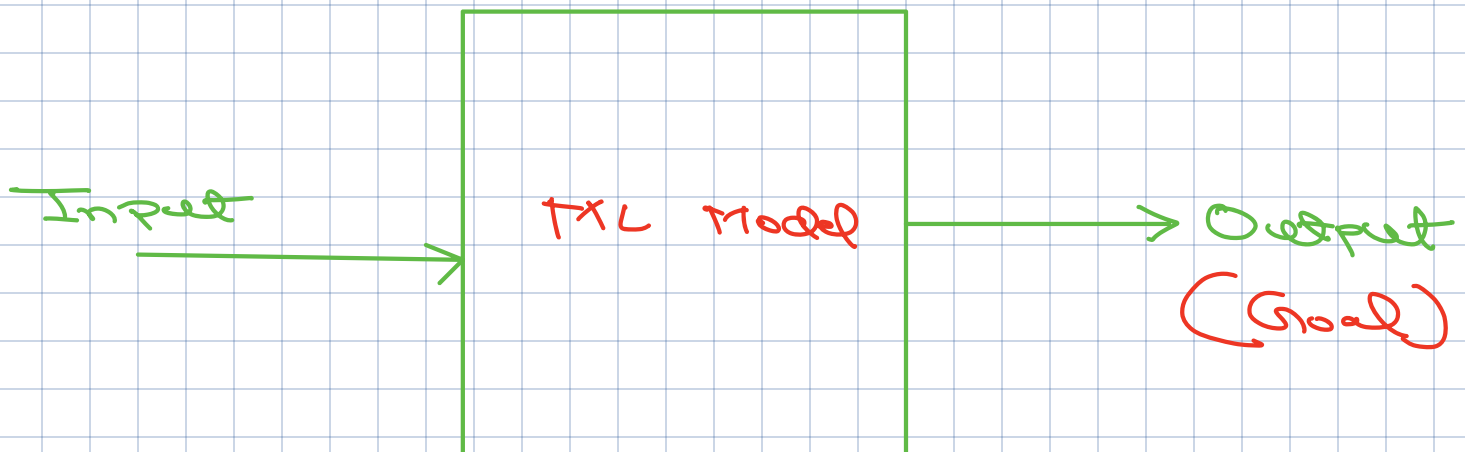


Agenda

- * Features vs Target
- * Feature Engineering
- * Skewness
- * Kurtosis
- * Univariate Analysis
- * Continuous to Categorical
- * Creating and Testing New features



Features vs Targets

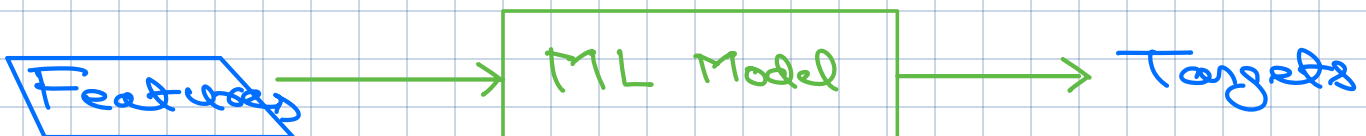
What is a feature?

Customer Dataset

index	education	Gender	Income	fitness usage	Product
0	10 th	M	20,000		P ₁
1	12 th	F	30,000		P ₁
2	Masters	M	60,000		P ₃
..	..	..	..		..
..	..	..	..		..
..	..	..	..		..
..	..	..	..		..
n					

Goal: Given 'details' of a person build ML 'Model' to Recommend Best Product.

Features	Targets
Input to our ML Model	Output of your ML Model



Feature Engineering

Garbage In $\longrightarrow$ Garbage Out

Processing Existing Features to create new more Informative Features

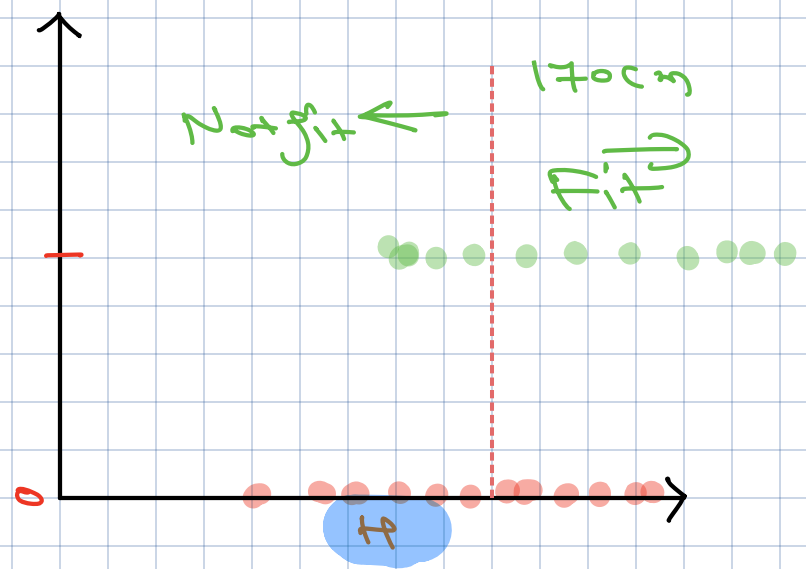
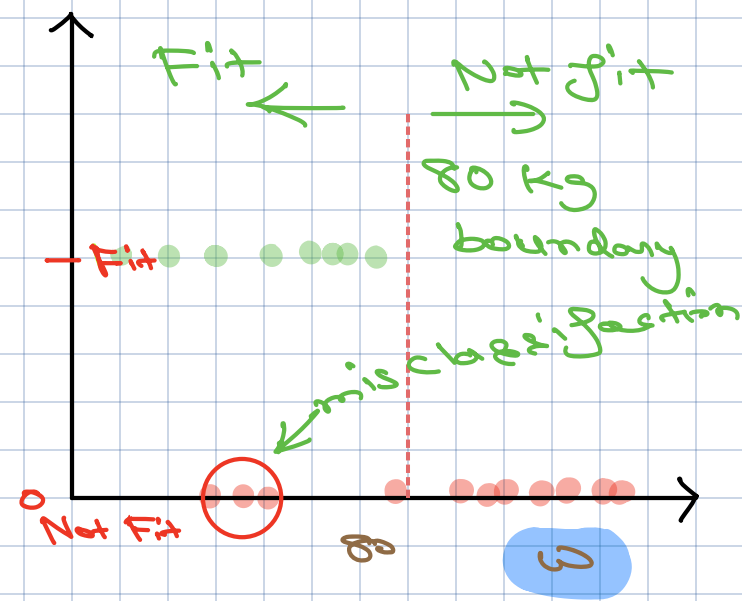
Ex: Given weight and Height of a person predict if they are Fit or Not Fit

H	W	Fit
170	70	Yes
160	65	Yes
165	80	No
150	90	No
''	''	''
''	''	''

Features = H and W

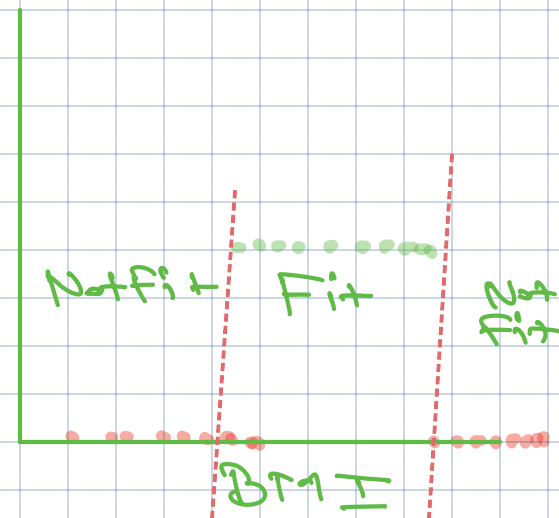
Target = Fit $\begin{cases} \text{Yes} \\ \text{No} \end{cases}$

Can we plot the features against Targets



Q Is there anything that we can do to separate Fit and Non-Fit people cleanly?

H	W	BMI	Fit
170	70	W/H	Yes
160	65	W/H	Yes
165	80	"	No
180	90	"	No
"	"	"	"
"	"	"	"



Conclusion

Loan Status Case-study

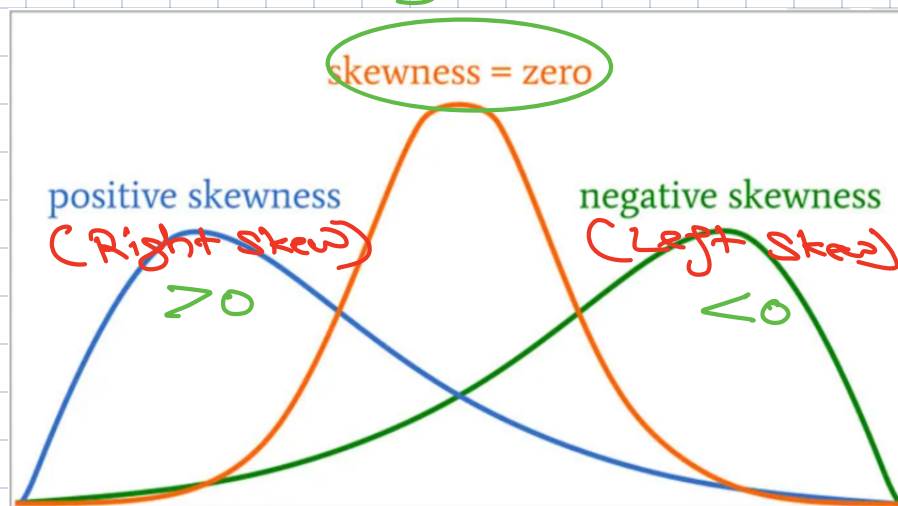
Goal: Develop a ML Classification model that uses Customer Data to predict if Loan should be approved or Not

index	education	Gender	Income		Loan Status
0	10 th	M	20,000	-----	
1	12 th	F	30,000	-----	
2	Masters	M	60,000	-----	
''	''	''	''	-----	
''	''	''	''	-----	
''	''	''	''	-----	
''	''	''	''	-----	
n					

① Skewness
② Kurtosis } Characteristics of Numerical Data

Skewness

Skewness is a measure of Asymmetry in the distribution of Numerical Data



* No Skewness (Symmetric)

- 1 Data is symmetrical
- 2 Mean $\approx$ median $\approx$ Mode

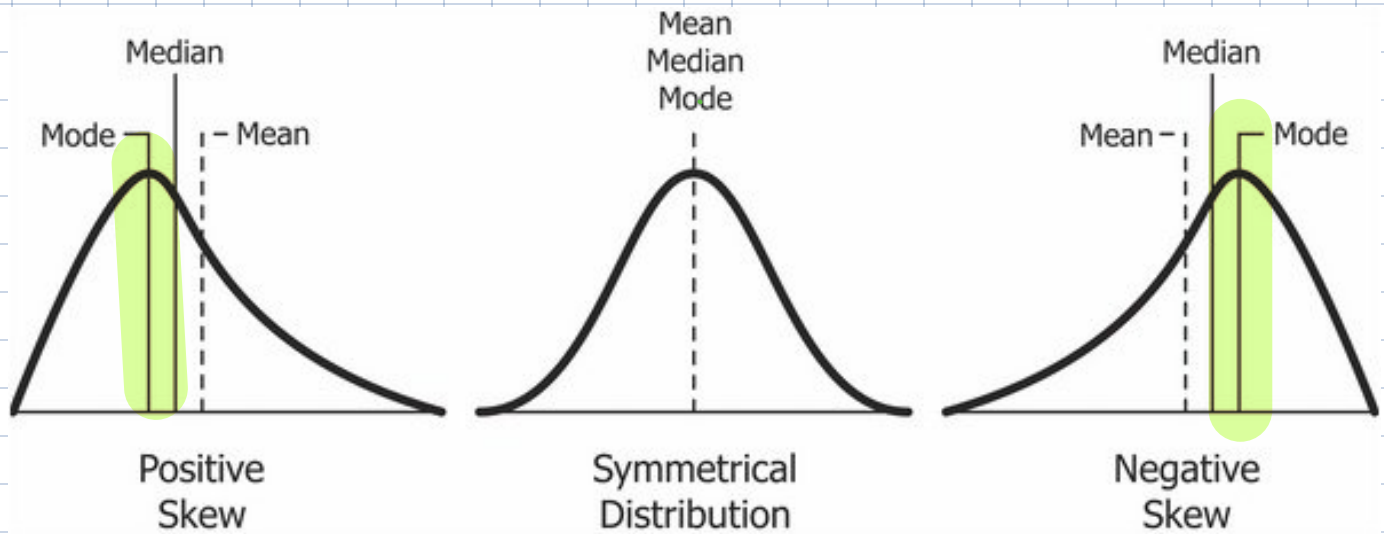
* Positive Skew (Right Skewed)

- 1 There are significant Number of Data points on the Right Hand Side of Peak (Right tail)
- 2 There can be potential Outliers

* Negative Skew (Left skewed)

- 1 There are significant Number of Data points on the Left Hand Side of Peak (Left tail)
- 2 There can be potential Outliers

Mean vs Median vs Mode



Post-Read

The skewness (g_1) of a dataset can be calculated using the formula:

$$g_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Where:

- n = Number of observations
- x_i = Individual data point
- $\bar{x}$ = Mean of the data
- s = Standard deviation of the data

Alternatively, **Pearson's moment coefficient of skewness** can be simplified as:

$$\text{Skewness} = \frac{\text{Mean} - \text{Median}}{\text{Standard Deviation}}$$

Interpretation of Skewness

Skewness > 0 Right Skewed (Positive)

Skewness < 0 Left Skewed (Negative)

Skewness $= 0$ No Skew (Symmetric)

Kurtosis

Helps us identify the thickness of tails

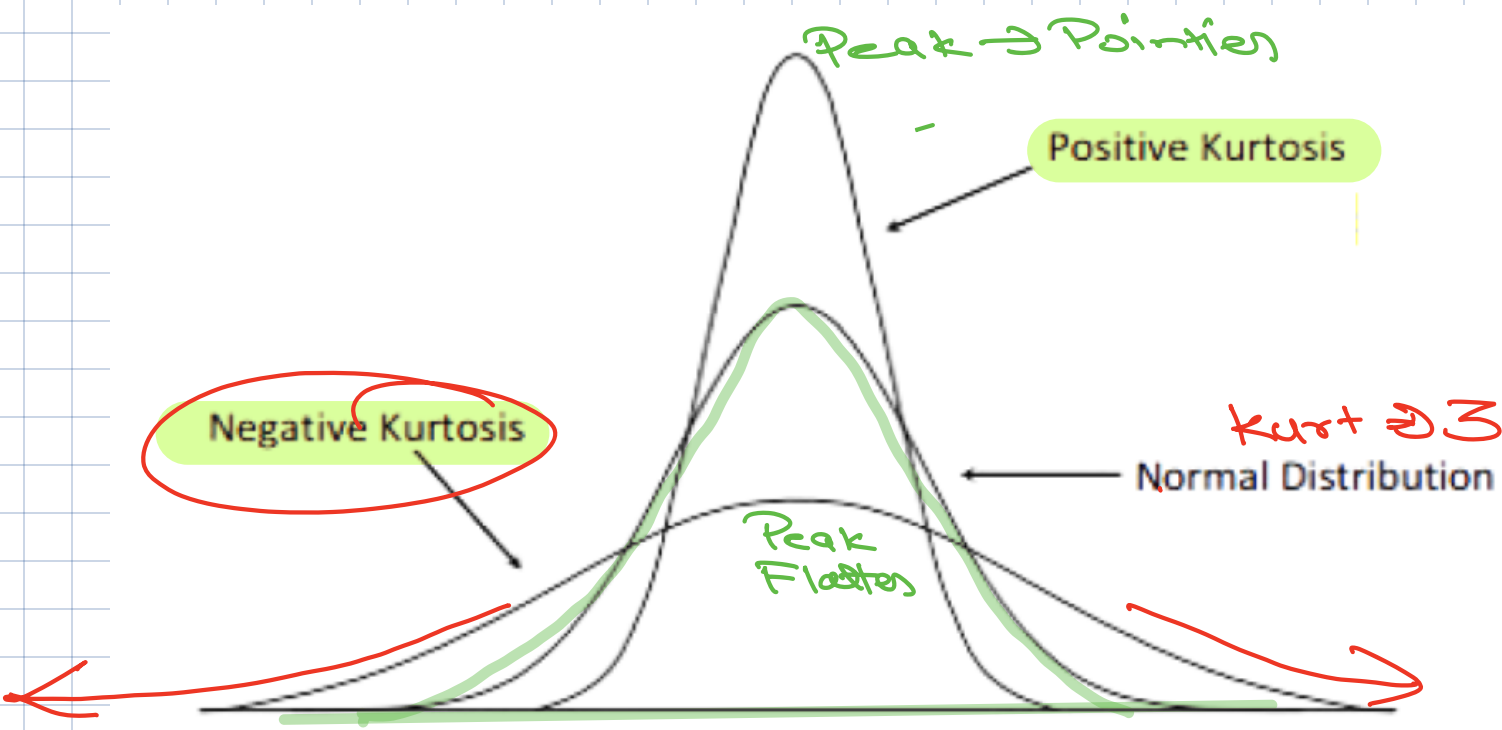
Kurtosis measure 'Sharpness of Peak' of Data Distribution

High kurtosis	Low kurtosis

Excess Kurtosis

$$\text{Kurt data} - \text{Kurt Normal}$$

$$\text{Kurt data} - 3$$



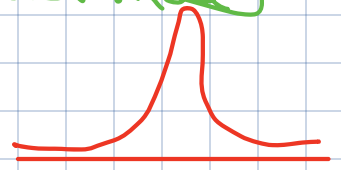
$$Excess_k = Kurt_{dist} - 3$$

① Leptokurtic (Pointier than Normal)

① Kurtosis > 3

① $Ex_k > 0$

① They Have Longer tails and Higher chances of outliers on both ends

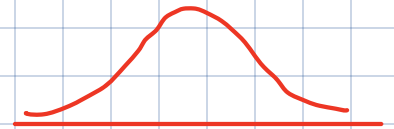


② Mesokurtic

① Kurtosis $= 3$

① $Ex_k = 0$

① Nearly Normal

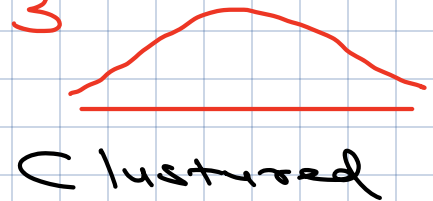


③ Platykurtic

① Kurtosis < 3

① $Ex_k < 0$

① Most DP are around Mean



What is the relationship between skewness and

Excess kurtosis in a normal distribution?

4 options

Active Duration (Most preferred: 30 seconds)

Appears for

60 Secs

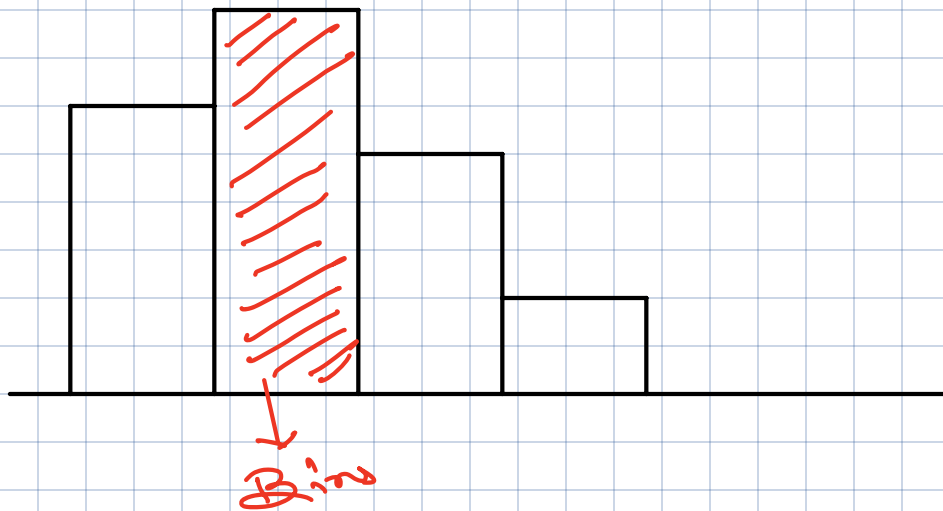


$Ex_k = 0$

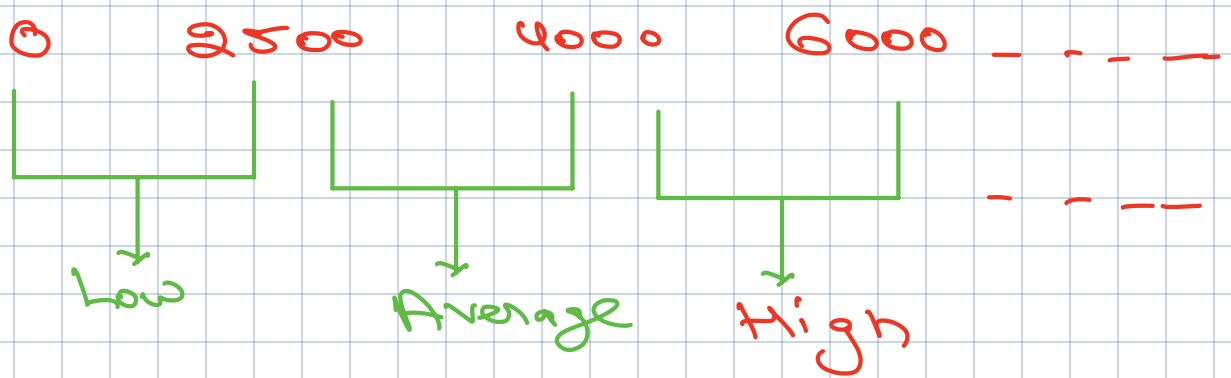
Skewness $= 0$

New Feature

Binning



Define interval $\longrightarrow$ Categories (Bins)



Co-app	Cat
1500	Low
2000	Low
3502	Avg
99	99

Feature: **Able to Pay EMI**

Consider Following two Loan applicants

- ① P1 earns 20L and applies for 50L Loan
- ② P2 earns 30L and applies for 5CR Loan

Who is more Likely to Get Loan?

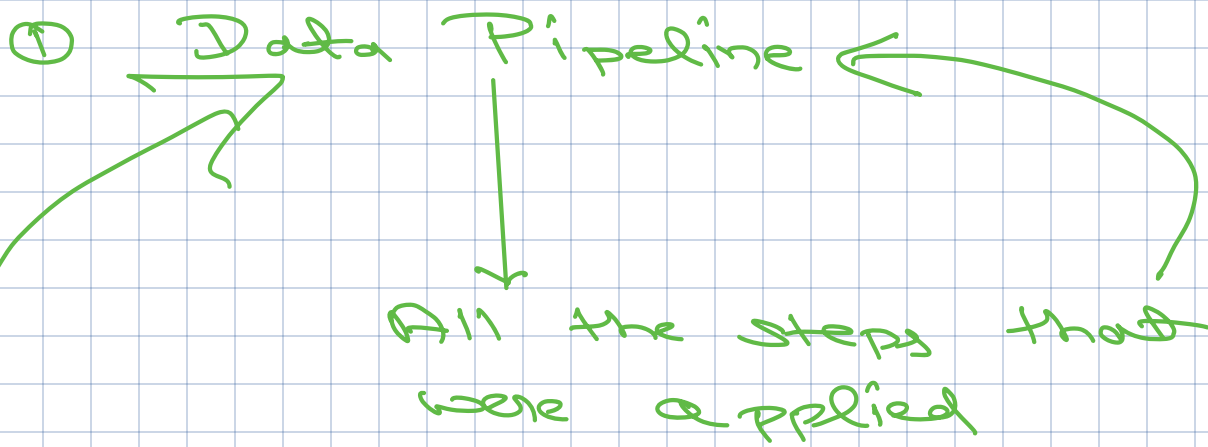
- * Loan amount term $\Rightarrow$ Num-months
- * Loan Amount $\Rightarrow$ \$ Value
- * Salary $\Rightarrow$ \$ Value

* 30% rule

EMI $>$ 30% of your monthly Income

Able-to-pay-EMI $\begin{cases} \text{Yes} \\ \text{No} \end{cases}$

$$\text{monthly EMI} \Rightarrow \frac{\text{Loan Amount}}{\text{Loan Term in Months}}$$



② Monitoring

Data Drift

① If distribution is similar for Numerical

② Categories are Expected

Alert

```
graph TD; A[② Monitoring] --> B[Data Drift]; B --> C[① If distribution is similar for Numerical]; B --> D[② Categories are Expected]; C --> E((Alert)); D --> E; E --> F[① Data Pipeline];
```

Data T1 L Modeling